11 January 2021

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Abstract

Providing IUPAC chemical names is necessary for chemical information exchange. We developed a Transformer-based artificial neural architecture to translate between SMILES and IUPAC chemical notations: Struct2IUPAC and IUPAC2Struct. Our models demonstrated the performance that is comparable to rule-based solutions. We proved that both accuracy, speed of computations, and the model's robustness allow us to use it in production. Our showcase demonstrates that a neural-based solution can encourage rapid development keeping the same performance. We believe that our findings will inspire other developers to reduce development costs by replacing complex rule-based solutions with neural-based ones. The demonstration of Struct2IUPAC model is available online on Syntelly platform https://app.syntelly.com/smiles2iupac

Keywords

machine learning, chemical nomenclature, Artificial Intelligence, Transformer, IUPAC Nomenclature

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Struct2IUPAC – Transformer-based Artificial Neural Network for the Conversion Between Chemical Notations

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Abstract

Providing IUPAC chemical names is necessary for chemical information exchange. We developed a Transformer-based artificial neural architecture to translate between SMILES and IUPAC chemical notations: *Struct2IUPAC* and *IUPAC2Struct*. Our models demonstrated the performance that is comparable to rule-based solutions. We proved that both accuracy, speed of computations, and the model's robustness allow us to use it in production. Our showcase demonstrates that a neural-based solution can encourage rapid development keeping the same performance. We believe that our findings will inspire other developers to reduce development costs by replacing complex rule-based solutions with neural-based ones.

Introduction

Before the Information Age chemical names were the universal language for the description of chemical structures. In the infancy of organic chemistry, there were no common rules for the naming of chemical compounds. However, the extensive growth of explored chemical space in the XIX century motivated chemists to harmonize chemical naming globally. In 1919 International Union of Pure and Applied Chemistry (IUPAC) was founded, and this non-commercial organization manages the development of chemical nomenclature. IUPAC publishes the Nomenclature of Organic Chemistry, commonly known as the "Blue Book." [1] The "Blue Book" provides guidelines to give unambiguous names for chemical compounds.

Nowadays there are several alternative representations for organic structures. For example, Simplified Molecular Input Line Entry System (SMILES) was designed to be handy both for humans and computational processing. However, IUPAC names still play an important role in organic chemistry. They are obligatory for processing chemicals in many regulated protocols, for example, for REACH registration in the EU, patent application submission in many countries, regulatory submission to FDA in the US. Most chemical journals require IUPAC names for organic structures too. Chemists are simply used to them. That is obvious that IUPAC names do not sink into oblivion in the nearest future.

In the past, chemists created IUPAC names manually. This process was error-prone because it requires deep knowledge of the nomenclature and attention. [2] It is hard for humans to perform the naming process accurately because it is rather algorithmic by nature. Moreover, chemists are biased towards trivial names and do not want to discard the names that they are used to utilize. Computers alleviate this problem. Now chemists use software for the name generation widely.

The history of names generators begins from the pioneering Garfield's work. [3] However, the first "everyday use" software for chemists was created and popularised only at the end of the XX century. Now, there are several commercial programs for generating IUPAC names: ACD/Labs, ChemDraw, Marvin, IMnova IUPAC Name, etc. Also, there is a framework Lexi-

Chem TK that provides an application programming interface (API) for some programming languages.[4] Nevertheless, there is no an open-source tool for the structure-to-name translation. Licensing agreements with the existing solutions, like ChemDraw JS and LexiChem TK, do not allow embedding to other platforms.

Amazingly, there is an open-source tool for the name-to-structure translation: OPSIN developed by Daniel Lowe.[5] But, as we mentioned above, there is no one for the reverse problem: structure-to-name conversion.

Recurrent neural networks and Transformer have been successfully used for natural language translation.[6,7] It is worth mentioning that recently a neural model for direct translation from English to Chinese languages was proposed.[8] However, the possibility of direct conversion from a structural representation to chemical names has not been properly discussed yet.

Our work was motivated by a very practical request – we had needed a tool for structure-to-name translation. We estimated the development costs of such a tool "from scratch" as unacceptable. Instead, we built a Transformer-based neural network that can convert molecules from SMILES representations to IUPAC names and back. In this paper, we describe our solutions, discuss its advantages and disadvantages, and show that Transformer can provide something that resembles human's chemical intuition.

Materials and Methods

Database

Deep learning requires large amount of data. PubChem is a large freely-available collection of chemical compounds with annotations.[9] We used chemical structures and corresponding IUPAC names from this database. It had 94 726 085 structures in total. The processing and training on the full PubChem database is time expensive, and about 50M samples seems more than enough, so we splitted the database into two parts and used one half for training

and the other one for testing. Structures that can not be processed by rdkit were removed resulting in 47 312 235 structures in the training set and 47 413 850 in the test set.

IUPAC and SMILES tokenizers

Tokenization – is a process of splitting sequences into pieces and demarking such pieces (tokens). It is a common preprocessing stage for language models. We use simple character-based SMILES tokenization and implemented a rules-based IUPAC tokenizer (*Fig. 1*). Our IUPAC tokenizer was manually designed and curated. We collected all suffixes (-one, -al, -ol, etc.), prefixes (-oxy, -hydroxy, -di, -tri, -tetra, etc.), trivial names (naphthalene, pyrazole, pyran, adamantane, etc.) and special symbols, numbers, stereochemical designations ((,), [,], -, N, R(S), E(Z), λ, etc.). Our tokenizer was able to correctly process more than 99% of molecules from PubChem[1].

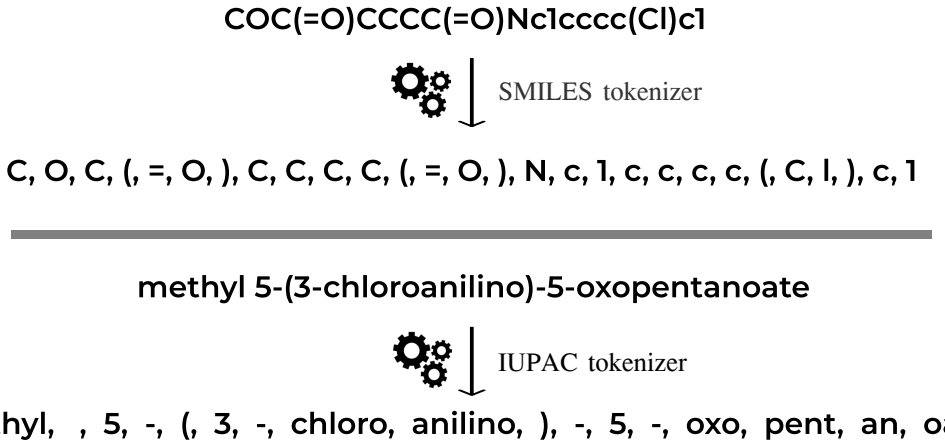


Figure 1: Demonstration of SMILES tokenization (top) and IUPAC names tokenization (bottom)

[1]Due to a technical mistake, we did not include some tokens in the IUPAC tokenizer (for example, "selena"). This makes it impossible to process molecules containing these tokens. We excluded from the train and test sets all molecules that cannot be correctly tokenized. Since retraining Transformer is computationally intensive, we have not fixed it yet.

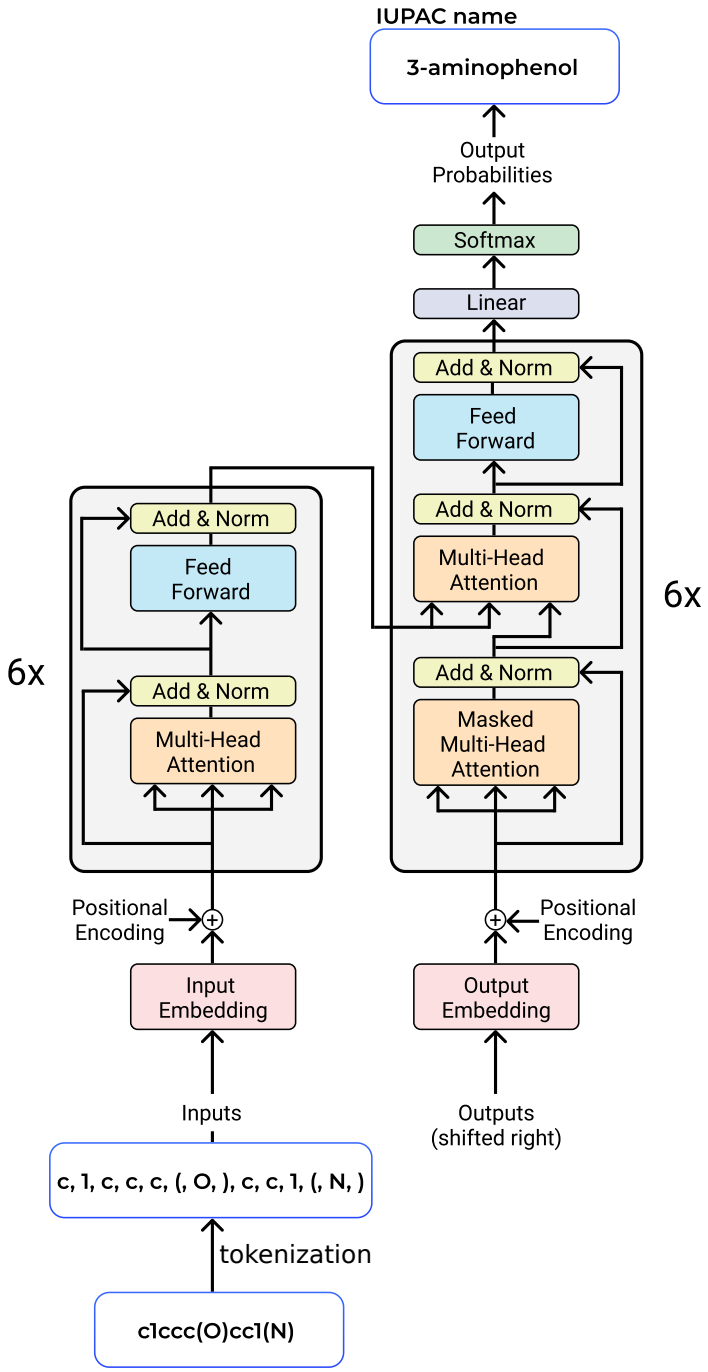


Figure 2: A scheme of *Struct2IUPAC* Transformer. Adopted from[6]

Transformer model

Transformer is a modern neural architecture designed by Google team, especially to boost the quality of machine translation.[6] Since its origin, Transformer based networks notably boosted the performance on NLP problems leading to newsworthy GPT models.[10] Transformer has been successfully applied to chemical-related problem: the prediction of outcomes of organic reactions[11] , QSAR modelling[12] and creation of molecular representations.[13] We used standard Transformer architecture with 6 encoder and decoder layers, 8 attention heads. The attention dimension was 512, and the dimension of the feed-forward layer was 2048. We trained two models: *Struct2IUPAC* that converse SMILES strings to IUPAC names and *IUPAC2Srtuct* – that performs reverse operation. There was no ultimate practical use for *IUPAC2Srtuct* model because an open-source OPSIN can be successfully used instead. However, to study reverse converter performance and following the aesthetic symmetry principle, we created these two models. The schema of our *Struct2IUPAC* model is given on *Fig. 2*

Production mode

Our scheme involves artificial neural networks, that means the solution is probabilistic by nature anyway. However – the generation of a chemical name is a precise task: a name can be either correct or wrong. We believe that the denial of translation is better than false conversion. Transformer can generate several versions of a sequence using beam search. By using OPSIN we can validate generated chemical names to guarantee that this name corresponds to the correct structure. So we can detect failures of our generator and do not display the wrong name.

Results and discussion

In order to validate the quality of our models we sampled randomly 100 000 molecules from the test set and calculated the percentage of correct predictions with different beam size.

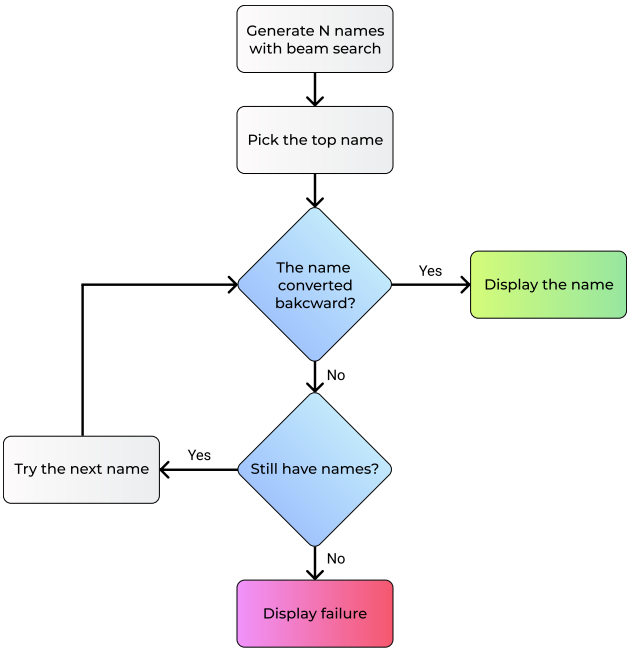


Figure 3: A scheme of "Production mode"

Our SMILES to IUPAC names converter, running in production mode with beam size = 5, achieved 98.9% accuracy (1075 mistakes per 100 000 molecules) on a subset of 100 000 random molecules from the test set. Transformer demonstrates the ability for the precise solution of an algorithmic problem, and that fact raises a new paradigm in software development. Before that there was a consensus opinion that artificial neural networks should not be used if a precise (strict algorithmic) solution is possible. Meanwhile, our approach is built on top of typical neural architecture and does not require algorithmic coding and chemical rules collection. The implementation of our system required about one and a half employee months for the whole pipeline. It is hard to estimate the resources that one needs to develop an algorithmic-based generator with competitive performance. Our preliminary estimation about the development of IUPAC names generator "from scratch," even using the source of OPSIN, would take more than a year for a small team. Anyway, we did not quantify our potential expenses, so we prefer to leave this question for a reader's opinion. Also, we believe, that our approach can be even more helpful for legacy data. It happens that there is a lack of documentation for legacy technical notations within the presence of some coincide

data. Engineers have to perform "Rosetta Stone investigations" before implementing the converter. One has to spend double efforts for such cases. In our approach, a neural network solves both tasks saving developers time.

It is known that Transformer fails at large sequences. Moreover, a common issue related to Transformer is a weakness in the processing of extra short sequences. For example, testing the model manually, we found a problematic molecule: methane. To estimate the applicability domain of our model we took 1000 examples for each length from 3 to 10 with a step of 1, from 10 to 100 with a step of 5 and 100 to 300 with a step of 10. As a result, we found that our model achieves accuracy close to 100% in the interval from 10 to 60 SMILES tokens. The result of the experiment is given on *Fig. 4*.

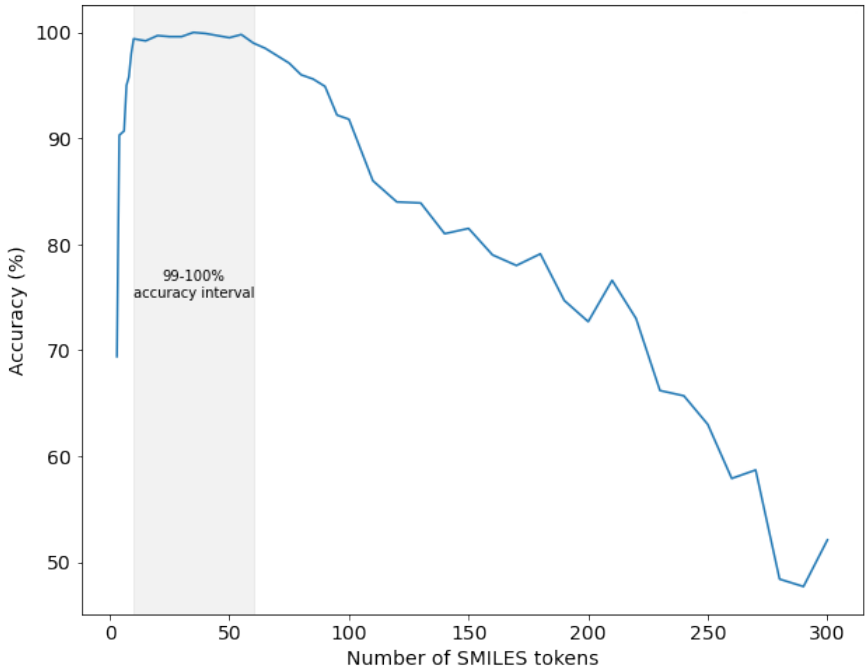


Figure 4: The dependence between model accuracy and the length of SMILES

We also depicted the distribution of sequence lengths on the test set (*Fig. 5*). The mean value of SMILES length is 46,0 tokens and IUPAC length is 40,7 tokens. So the majority of the PubChem molecules is within the applicability domain of our model.

We compared our IUPAC to SMILES Transformer model (*IUPAC2Struct*) with the rules-based tool OPSIN on the test set (*Table. 1*). Our converter achieved 99.1% accuracy (916

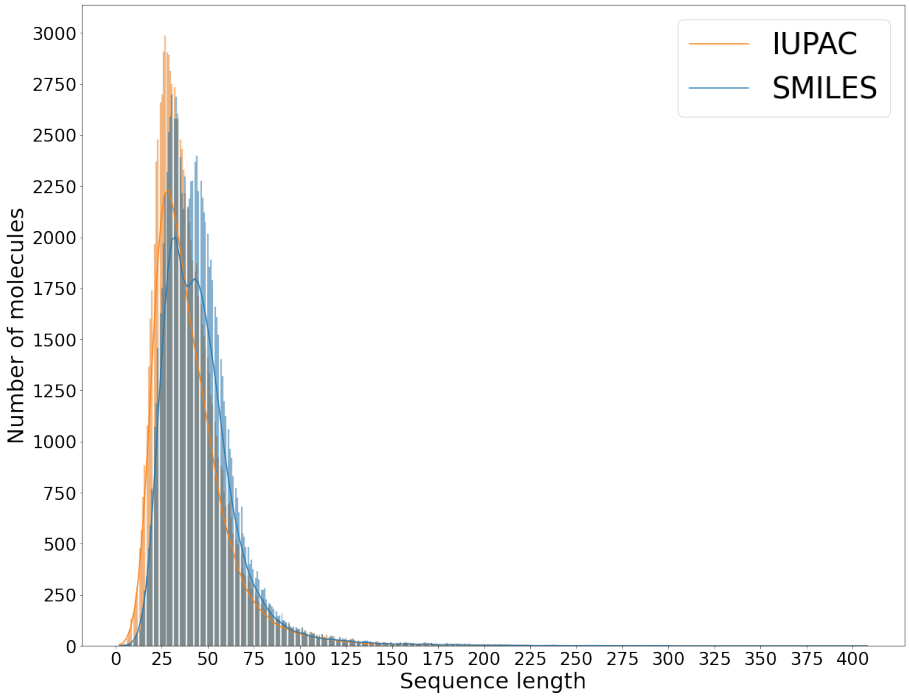


Figure 5: The distribution of the lengths of SMILES and IUPAC on the test set

mistakes per 100 000 molecules) and OPSIN performed 99.4% (645 mistakes per 100 000 molecules).

Table 1: Accuracy (%) of models on the test set of 100k molecules with different beam size

Struct2IUPAC			*IUPAC2Struct*			OPSIN
Beam 1	Beam 3	Beam 5	Beam 1	Beam 3	Beam 5	
96.1	98.2	98.9	96.6	98.6	99.1	99.4

Transformer is a heavy neural architecture. The application of Transformer can be slower than for algorithmic-based solutions. To understand this model's practical applicability in terms of execution time, we estimated the speed of name generation both on CPU and GPU. We measured the dependence between the number of output IUPAC tokens and the time required for several Transformer runs without beam search and validation feedback loop. The result of the experiment is given on *Fig. 6*. Transformer consists of two parts: encoder and decoder. Encoder runs only once to read SMILES input, whereas decoder processes each output token. For this reason, only the output sequence length influences the time of execution. One can see that GPU is notably faster than CPU. GPU application requires less than 0.5 seconds even for chemical names with maximal length. This time-frame is acceptable for the practical usage.

Our solution requires prominent computational resources to train the model. The production model has been trained for ten days on a machine with 4 Tesla V100 GPUs and 36 CPUs, with a full load. Still, nowadays human work hours are becoming more valuable, and machine hours cheaper and our approach meets this way.

The most intriguing ability of Transformer to operate with IUPAC nomenclature in a chemically reasonable way. One can see that the model can infer some chemical knowledge. For example for a molecule on *Fig.7* model generates four chemically correct names (OPSIN converts these IUPAC names to the same structures) :

- *3-[[2-[3-(tert-butylcarbamoyl)anilino]acetyl]amino]-N-(3-methoxypropyl)benzamide*
- *N-(3-methoxypropyl)-3-[[2-[3-(tert-butylcarbamoyl)anilino]acetyl]amino]benzamide*

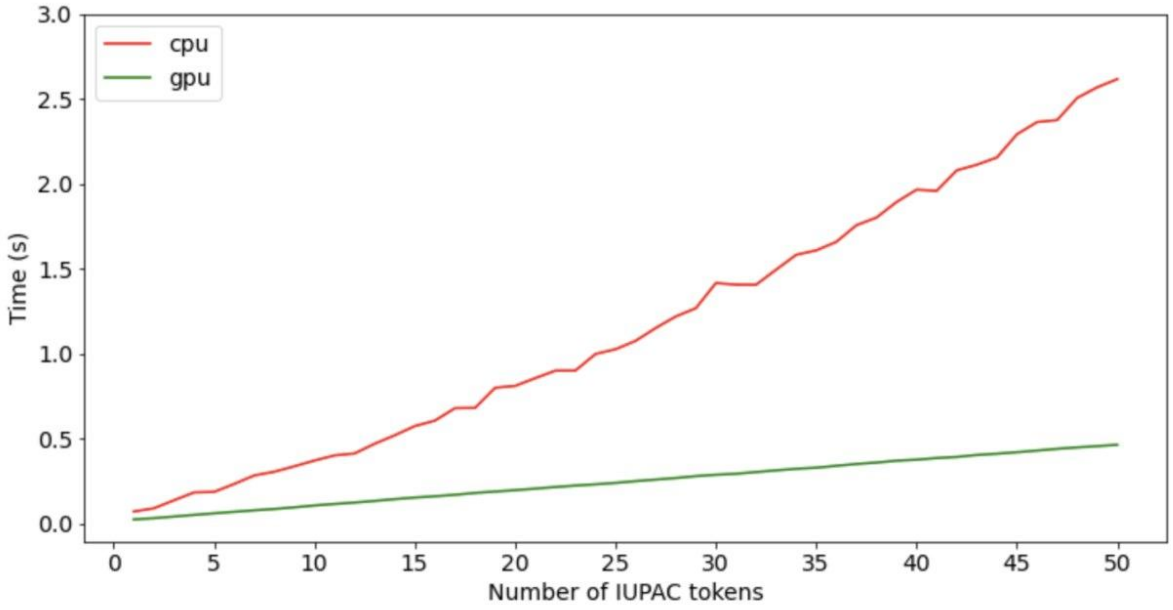


Figure 6: The correlation of mean time and output sequence length.

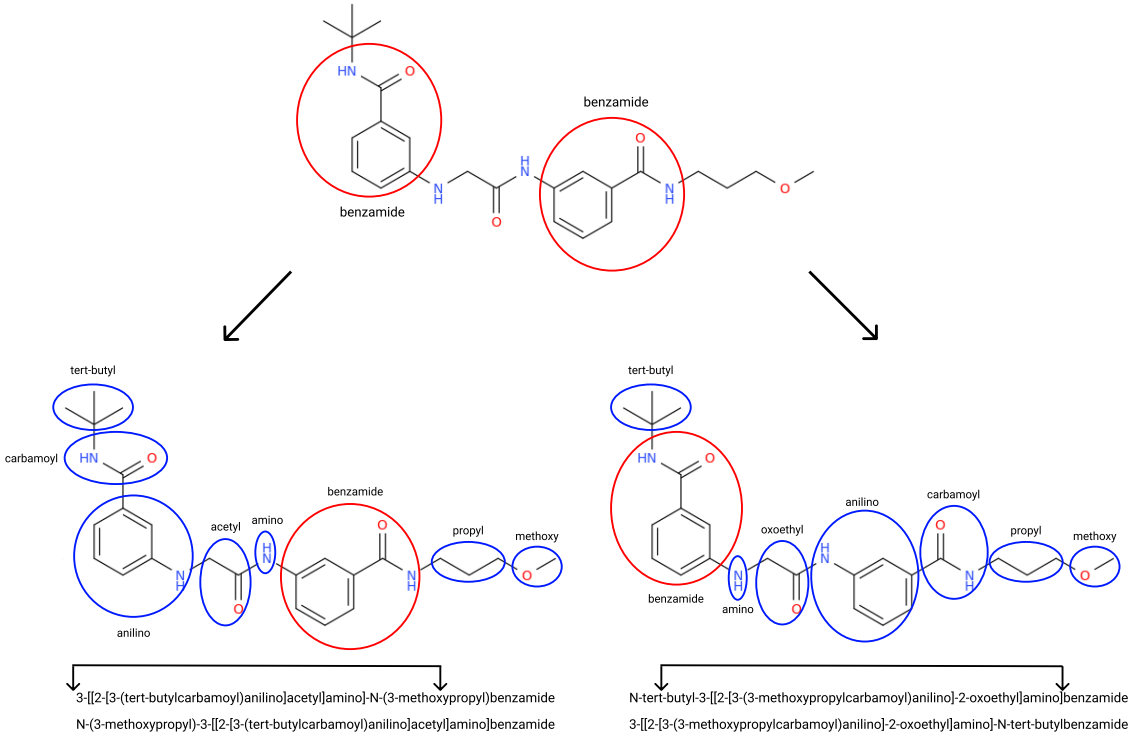
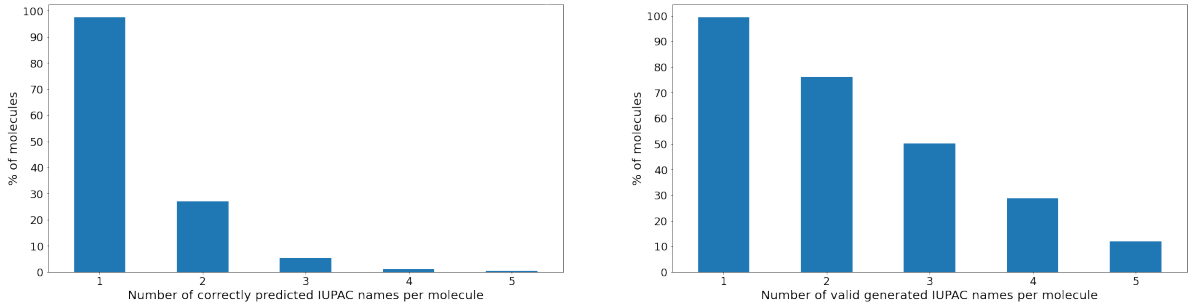


Figure 7: An example of a molecule with four correctly generated IUPAC names

- *N-tert-butyl-3-[[2-[3-(3-methoxypropylcarbamoyl)anilino]-2-oxoethyl]amino]benzamide*
- *3-[[2-[3-(3-methoxypropylcarbamoyl)anilino]-2-oxoethyl]amino]-N-tert-butylbenzamide*

This is due to the fact that the molecule has two parent benzamide groups (in red on *Fig.7*). Depending on the choice of the parent group, there are two different ways to name the molecule. Also, for each of the two methods, there are two more ways associated with the order of enumeration of substituents. As a result, we have four correct IUPAC names for one molecule, and all of these variants are generated by Transformer.

Been intrigued by this observation we analyzed the distribution of valid and correct molecules in a batch. We randomly sampled 1000 molecules from our 100 000 test subset and calculated: a) the number of true (correct) IUPAC names (reverse translation by OPSIN leads to the same molecule) b) the number of chemically valid names (OPSIN can process a molecule, however there is no guarantee that the molecule is the same). These distributions are given on *Fig. 8*. One can see that Transformer can generate up to two correct names for about 20% of molecules, and about 1% of molecules can have up to 4 correct IUPAC names. It supports our claims that Transformer does not just memorize common patterns but infers the logic behind IUPAC nomenclature.



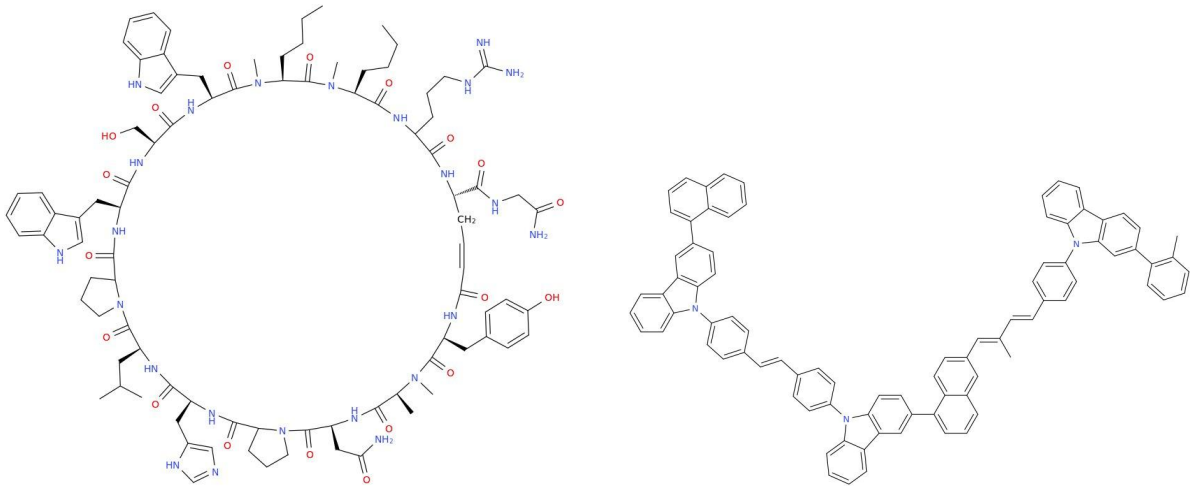
(a) Correct IUPAC names (b) Valid IUPAC names

Figure 8: The distribution of the number of names variations using a Transformer's beam search

It is interesting to track the behavior of Transformer outside the applicability domain. Our observations revealed that the performance of Transformer drops down with huge

molecules. In the range from 200 to 300 tokens there are two common types of mistakes. The first one is the situation when the model loses the opening or closing squared bracket. It fails the whole structure due to a lexical mistake. That means that the model is under-trained on such an extra-size dataset. This behavior was expected because there were small amount of very large molecules in the training set. The second typical case is losing a part of a large molecule. In this case, Transformer generates a chemically valid molecule, albeit that is shorter than the original. However, Transformer-based models are known for ability to work with thousands-long sequences, and we believe, that with enough large samples in a dataset, Transformer can achieve good performance on extra large molecules too.

Despite the fact that the accuracy of the model does not exceed 50% on very large molecules, we found the interesting examples of complex molecules for which IUPAC names were correctly generated (*Fig. 9*).



(a) A cyclic peptide (b) A carbazole derivative

Figure 9: Two examples of challenging molecules for which Transformer generates correct names

Conclusions

In this paper, we propose the Transformer-based solution for the generation of IUPAC chemical names using SMILES string. This model achieved 98.9% accuracy on the test set from PubChem. Also, the model reached close to 100% accuracy within 10 to 60 SMILES length. Our reverse model (*IUPAC2Struct*) reached accuracy 99.1%, comparable to open-source OPSIN software. We demonstrated that the computation time is generally applicable for production usage. We showed that our model operates well within a wide range of molecules length. Our research opens doors for a new paradigm in software development. We demonstrated that one could even replace a complex rule-based solution with modern "heavy" neural architectures. We believe that neural networks can now solve a wide range of so-called "exact" problems (problems for which an exact algorithm or solution exists) with compatible performance. We ask researchers to validate this idea for other algorithmic-based challenges.

Conflict of Interest

The authors declare the following competing interests: Maxim Fedorov and Sergey Sosnin are co-founders of Syntelly LLC. Lev Krasnov and Ivan Khokhlov are employees of Syntelly LLC. The authors are going to integrate the functionality described in this paper to Syntelly online platform.

Data Availability

The data is located on Zenodo (https://doi.org/10.5281/zenodo.4280814). It contains a subset of 100 000 chemical compounds that were used for testing Transformer, a subset of compounds on which OPSIN fails and compounds on which our *IUPAC2Smiles* model fails. Also, the 100 000 subset is placed on GitHub repository: https://github.com/sergsb/IUPAC2Struct with the *IUPAC2Struct* Transformer code.

Model Availability

Our *Struct2IUPAC* model is available for the community on *Syntelly* platform: https://app.syntelly.com/smiles2iupac – text interface only for testing SMILES to IUPAC model, https://app.syntelly.com/individual – graphical user interface for the prediction of properties of organic compounds and IUPAC names. Transformer and pretrained *IUPAC2Struct* model are available on GitHub: https://github.com/sergsb/IUPAC2Struct

Contributions

L.K., I.K., and S.S. performed the experiments described in this paper and conducted the analyses. I.K. implemented Transformer and trained the models. M.V.F. and S.S. initially designed the study. S.S. and L.K. wrote the draft of the manuscript, I.K. and M.V.F. revised it. S.S. supervised the project. All authors read and approved the final paper.

Acknowledgement

The authors acknowledge the use of Skoltech's Zhores GPU cluster[14] for obtaining the results presented in this paper. The authors are thankful to Arkadiy Buianov for the help in creation of the web-demo.

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